
Unit – 1 : Session – 5 : SLO – 1 | Exokernel and Layered Operating Systems — Word Reassembly

I. Key Terms (reassembled)

1. EXOKERNEL — a minimalist kernel that exposes hardware directly to applications through library operating systems; applications control resource management themselves.
3. LAYERED STRUCTURE — the OS is divided into layers; each layer uses services only of the layer below it (layer 0 = hardware, top layer = user interface).
5. IPC (Inter-Process Communication) — minimal service kept inside small kernels.
6. LIBRARY OS — in exokernels, untrusted application libraries implement abstractions like files.
7. MODULARITY — layered design allows debugging and verification layer by layer.

II. Summary Answers

1. Exokernel: purpose is maximum flexibility and performance by multiplexing hardware securely and letting applications manage resources via library OSes (e.g. MIT's Aegis/XN).
3. Layered OS: architecture where functions are grouped into ordered layers; advantage is simplicity of construction/debugging; disadvantage is overhead of crossing layers and careful layer definition.